-14/12 ms (2)

Reg. No. | | | | | | | | | | | [17CS3IOT01]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2018

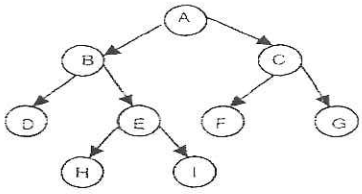
DATA STRUCTURES USING C

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing atleast two from each part.
ii. Missing data may be suitably assumed.

PART-A

1.	a.	Define Data structure. List and explain classification of Data structure.	07
	b.	Explain structure, unions and self-referential structure with example.	07
2.	a.	With an example explain pointer declaration, pointer initialization and use of pointer in dynamically allocated array.	07
	b.	Define dynamic memory allocation. Explain how memory can be allocated using dynamic memory functions.	07
3.	a.	Explain Insertion sort with an example and write C function for the same.	10
	b.	Discuss the difference between sequential and binary search.	04
4.	a.	Write an algorithm to evaluate postfix expression ,trace the same algorithm with stack contents for the following expression 651 -4 *23 ^ /+	07
	b.	Write a C function to implement push and pop function for stack using arrays.	07

PART-B

5.	a.	Discuss the advantage of circular queue over linear queue. Write C routines for inserting and deleting an element from circular queue.	07
	b.	Write a C function to implement enqueue and dequeue operations in a que using linked list.	07
6.	a.	Explain the different types of linked list with diagram.	10
	b.	Write the node structure for linked representation.	04
7.	a.	Define graph .Explain different representation of graph with example.	10
	b.	Define the following: i. Connected graph ii. Directed graph	04
8.	a.	Explain binary tree. How it is represented using array &link list.	07
	b.	Trace the preoder, postorder and inorder traversal for the given tree:	07